

Kim Personal Reflection

Structure and Goals

In the beginning of this course, I would say that I acted as a sort of leader for the rest of the group, and I made sure to allow everyone to voice their own ideas about the types of projects we would undertake throughout the course. For our AR project, we decided to create an image-tracked AR experience. I tried to convey to the group that we should keep the scope simple and focus on creating something that is actually showable, so we loaned a picture book from the library and started creating individual models for each picture. This was extremely simple, and it led to the group wondering what to do next. Because of that, we expanded the scope and decided to take a risk with Meshy.ai so that, instead of static models, we could capture a picture with our mobiles during runtime, send that picture to Meshy, and create an AR experience for any given image. This was extremely experimental and risky, and I undertook this stressful task myself with some assistance—creating the scripting pipeline and handling the texturing, which was a hassle. I'll talk more about this in a different section.

Work structure and Github

I would say that we worked well together, and we shared the same vision for the outcome of the two projects. In terms of how we worked together, we usually met up in the labs, loaned one of the computers, and worked side-by-side on the big screen. Whoever was controlling the big screen would then get support from the others using their own laptops, using Google and AI to troubleshoot the different challenges we encountered along the way. We also recognized that the person controlling the big screen would benefit the most in terms of learning. Because of that, we agreed that we would each create our own scenes within Unity while working from home, and whenever we met up again, we would cannibalise anything useful from that work into the main scene. This ensured that everyone contributed to the project in a meaningful way.

In addition, our devblogs are separated into their own corresponding repositories. Everything related to AR is kept within the AR repository, and everything related to VR is placed in the VR repository.

AR Project - Image Tracking with Meshy

Catching up from the beginning, I would say the most stressful aspect of working on this project was the fact that I didn't have an Android phone. It extremely inhibited my ability to troubleshoot from home, and even though Kasper said we could use the Unity simulator to test, it would sometimes bug out or flat-out not work. VIA also had some Android phones for student use, but they were old models with outdated software versions that couldn't be upgraded, which caused even more errors during runtime.

Besides that, the way I worked on this project meant relying on another group member's Android phone to continue development and debugging. The main challenge I faced was trying to get the textures to display for the generated models from Meshy, which took a considerable amount of time to get working. We agreed that I would focus on solving this while the team worked on the VR section. I eventually managed to get the textures working during the last lab

session, which felt like a huge weight lifted from my shoulders. I'll list everything I've personally learned so far while undertaking this project:

- Reference Image Library - Sprite 2D
- AR Components Before Meshy - Audio Source Component - Collider Component - Event Trigger Component
- After Meshy: Scripting structure clarity and understanding of the pipeline Unity Graphics Settings – ensuring GLTFUtility shaders are always included during runtime to fix textures

VR Project – Zombie Wave Game

When I said I focused almost all my time on the AR project, that wasn't entirely true. Together with the group, we created a new repository and the overall setup needed to begin coding the fundamental foundation we had discussed. We started by creating scripts for the wave logic, adding placeholder cylinders to represent zombies, and importing the gun and table models into the scene. The gun model import included several different texture types that we needed to slot in, such as normal maps, metallic maps, and roughness maps. Afterwards, we also worked on attaching the magazine to the gun and creating a muzzle from which the bullet projectiles would spawn.

Things I contributed to in this section:

- Scripting the first version of the wave implementation with the team
- Importing and assigning the different maps to the gun model with the team
- Scripting the first version of the muzzle and magazine snapping with the team
- Overall following along for the rest of the project, including sound effects, animations, blood effects, and the game-over screen

Conclusion

Overall, I'm happy with the outcome of the two projects, especially considering our time constraints. I'm very proud of the VR project, but the AR project, as I mentioned, is more experimental, and there's very little information available beyond the Meshy documentation for what we attempted. Perhaps in the future this type of technology will be more streamlined for our purposes. Overall, I feel like I've become a more capable person within Unity.